



ONLINE RETAIL CUSTOMER ANALYTICS

Turning Transactions Into Loyal Customers

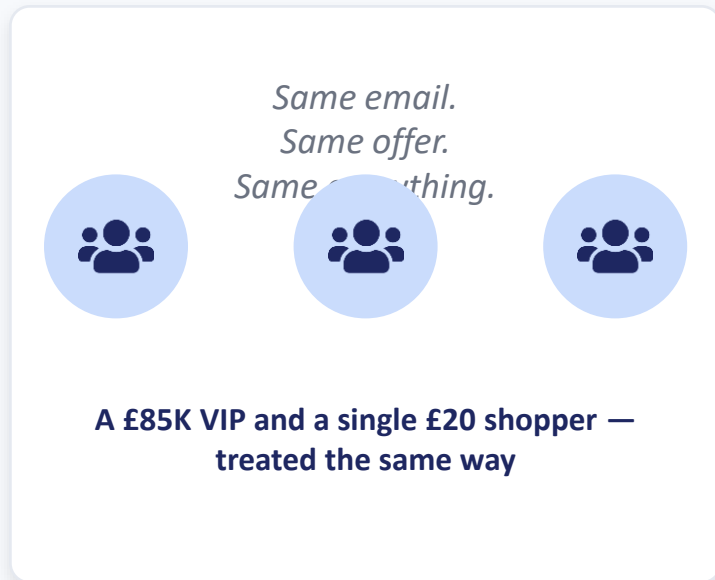
An RFM & K-Means Segmentation Study — UK Online Gift Retailer, 2010–2011

Presented by Engy Khaled

THE BUSINESS PROBLEM

One-Size-Fits-All Marketing Is Costing Us

- Every customer receives the same emails, offers, and attention — regardless of value
- Our biggest spenders and our one-time shoppers are treated identically
- Result: wasted budget on the wrong people, and missed chances to keep the right ones



From Messy Data to a Clean Foundation

- Removed rows with missing Customer IDs, duplicates, and cancelled orders
- Filtered out returns and non-positive quantities or prices
- Result: a clean, trustworthy dataset ready for analysis

541,909

raw transaction lines



392,692

clean, valid transactions

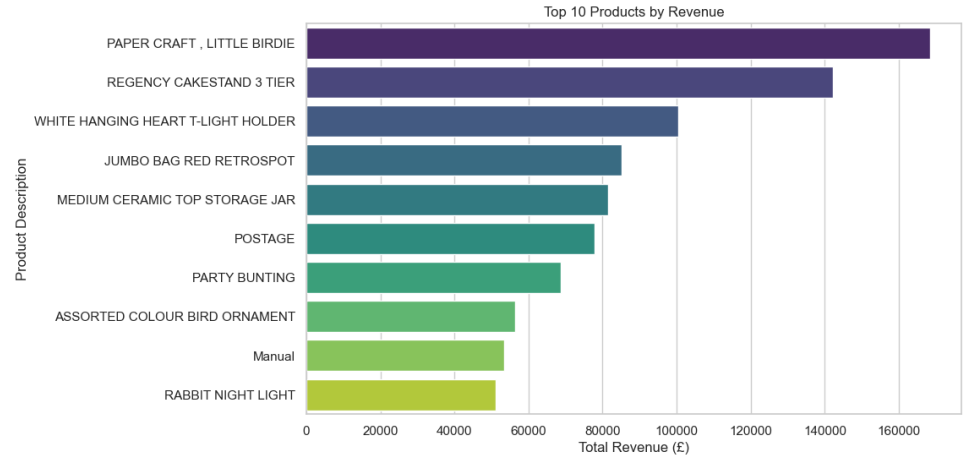
4,338

unique customers ready for analysis

KEY INSIGHT 1

A Few Products Carry the Business

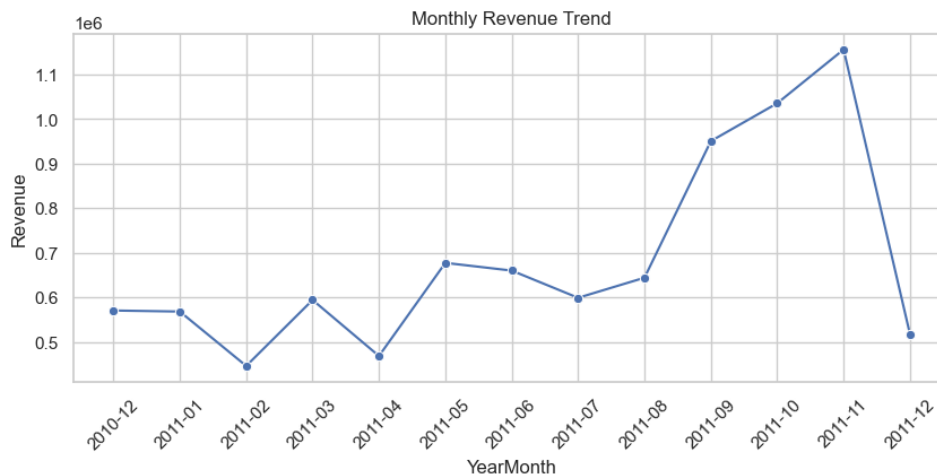
- Top 10 products generate roughly 20% of total revenue
- A textbook 80/20 (Pareto) pattern
- Stockouts on these bestsellers carry outsized revenue risk



KEY INSIGHT 2

November Is Our Make-or-Break Month

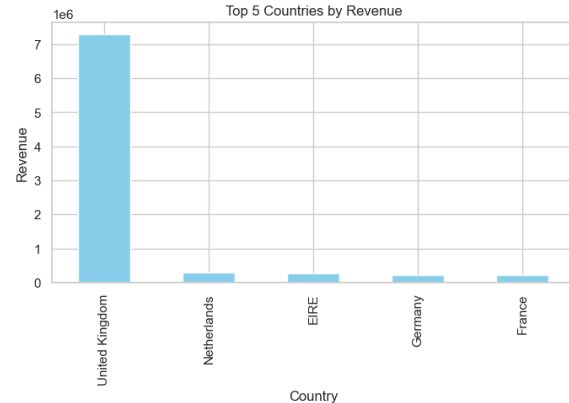
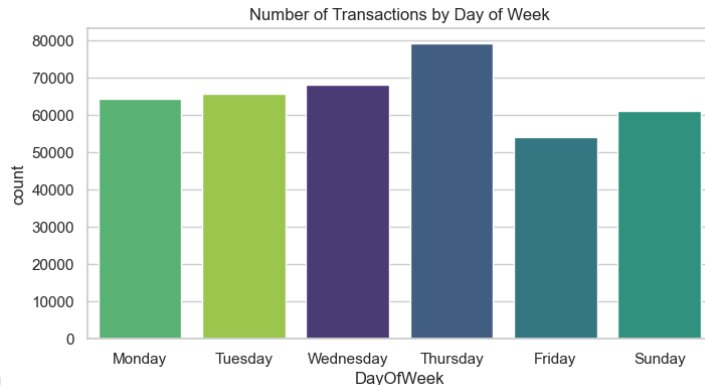
- Revenue climbs steadily through the year and spikes sharply in November
- Classic pre-holiday, Christmas gift-buying behavior
- Clear signal for inventory build-up, staffing, and campaign timing



KEY INSIGHT 3

When and Where Our Customers Shop

- Thursdays are peak trading days; the store sees no Saturday activity at all
- Purchases cluster around standard business hours, peaking near midday
- The UK dominates revenue — other markets remain under-tapped



The RFM Magic: Three Simple Questions

To understand any customer, we only need to ask three simple questions:



Recency

“How recently did they buy?”

Days since the last purchase. Sooner is better.



Frequency

“How often do they buy?”

Number of separate orders placed. More is better.



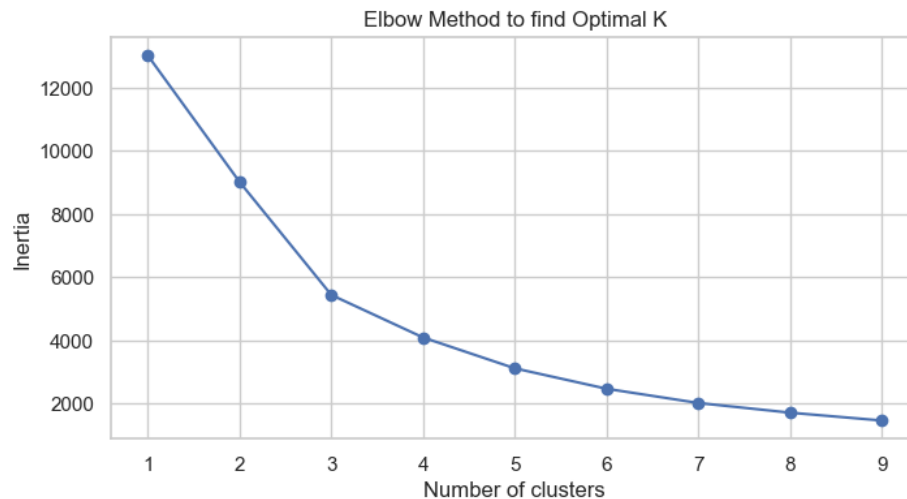
Monetary

“How much do they spend?”

Total revenue contributed. Higher is better.

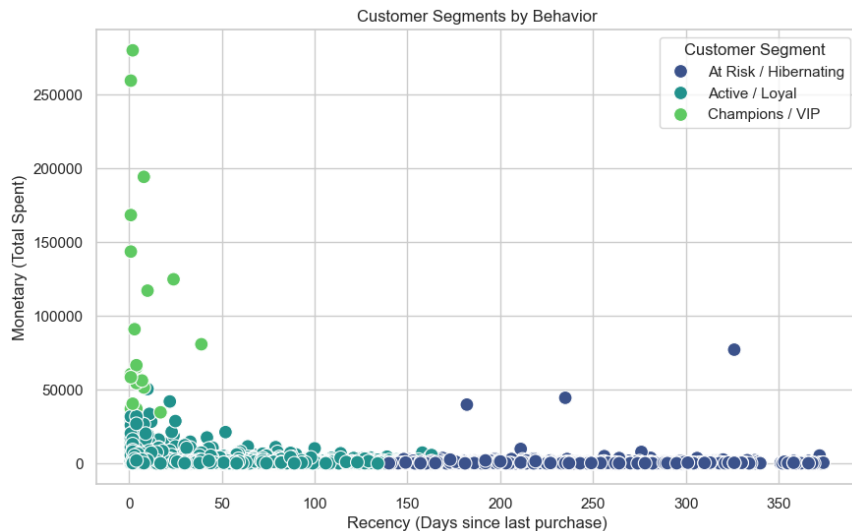
Finding the Right Number of Groups

- The Elbow Method shows a clear bend at 3 clusters
- A Silhouette Score of 0.594 confirms the groups are well-separated and cohesive
- Three segments balance simplicity with real business action



THE SEGMENTS

Meet Our Three Customer Segments



Champions

Avg. £85,826 spent · bought 6 days ago



Active

Avg. £1,850 spent · steady mid-value shoppers



At Risk

Avg. £630 spent · silent for 247+ days

What Marketing Should Do Tomorrow



Champions

Protect & Reward

- VIP loyalty tier & dedicated care
- Early access to new arrivals
- Referral rewards program



Active

Nurture & Grow

- Points-based loyalty program
- Cross-sell & upsell offers
- Milestone / anniversary perks



At Risk

Win Back

- “We miss you” email campaigns
- Time-limited reactivation offers
- Feedback surveys to learn why

From Raw Data to a Smarter Growth Strategy

392,692

clean transactions analyzed

3

validated customer segments

0.594

Silhouette Score (strong fit)

- A clean, statistically validated segmentation the marketing team can use starting now
- Protect Champions, grow Active shoppers, win back those At Risk
- A repeatable foundation for smarter CRM, inventory, and campaign decisions going forward



Thank You

Questions & Discussion